

Supplementary data

VVO - VHIO's Visual Omics

This document lists every panel available in the two VVO applications (Mutations and Bulk RNA-seq), and explains in simple terms what each panel shows and which settings the user can change. It complements Section 3 (Results) of the main text, which only describes the panels developed as part of this work.

1. Mutations application

The Mutations application shows which genes are mutated in each tumor sample, using an OncoPlot. The user uploads a variant table produced by ParseVCF, and can optionally add sample metadata and a reference cohort file for comparison.

1.1 Inputs (sidebar)

Before looking at the panels, it helps to know what can be uploaded and adjusted from the sidebar. These settings control which mutations are shown in the OncoPlot panel.

Required:

- Upload Excel (.xlsx): the variant table exported by ParseVCF.
- Variant calling mode: choose Paired (tumor + matched normal sample) or Tumor only, depending on how the file was generated.

Filters (applied to all mutations before drawing the plot):

- Filter column values: which FILTER values to keep (PASS is selected by default, if present).
- Min VAF tumor: minimum variant allele frequency required in the tumor sample.
- Max VAF control (Paired mode only): maximum variant allele frequency allowed in the normal sample.
- Min total reads tumor: minimum sequencing depth at the variant position.
- Min alternative reads tumor: minimum number of reads supporting the variant.
- Annotation impact: which SnpEff impact levels to include (HIGH / MODERATE / MODIFIER).
- CGI Oncogenic Summary (only shown if present in the file): filter by CGI oncogenicity category.
- OncoKB (only shown if present in the file): filter by Likely Oncogenic / Oncogenic.
- Mutation types to show: which variant classes to display (e.g. Missense, Nonsense, Frame Shift, etc.).

Optional inputs:

- Upload sample metadata (.xlsx): a table with clinical or technical information for each sample. Adds colored annotation tracks above the plot.
- Sample metadata columns to show: which of those columns to display (maximum 6).
- Reference cohort annotation file (cBioPortal format): a gene-level mutation frequency table from an external cohort, used for comparison.

Figure S1. Sidebar of the Mutations application, showing the required inputs, filters, and optional inputs described above, as displayed for an example uploaded dataset.

1.2 Panel: OncoPlot

This is the main panel of the Mutations application. It draws a grid where each column is a sample and each row is a gene. A colored square means that gene is mutated in that sample, the color tells you the type of mutation. A bar chart on top shows how often each sample is mutated, and a bar chart on the side shows how often each gene is mutated across all samples. If sample metadata was uploaded, colored bars below the plot show groups such as treatment or tissue type.

What you can change:

- Show genes mutated in at least (%): minimum percentage of samples in which a gene must be mutated to appear in the plot (maximum 75 genes are shown even if more genes pass this threshold).
- Add specific genes: lets the user manually force certain genes into the plot, even if they do not reach the percentage threshold.
- Gene order: only visible when a reference cohort file is loaded; sorts genes either by mutation frequency in the uploaded samples or by frequency in the external reference cohort.

Figure S2. Parameters inside Data Parameters of the OncoPlot Panel.

- ColData colors: lets the user change the color used for each sample metadata track.
- Gene name font size / Sample name font size: controls the text size of the row and column labels.
- Download PNG: exports the OncoPrint exactly as it is currently displayed.

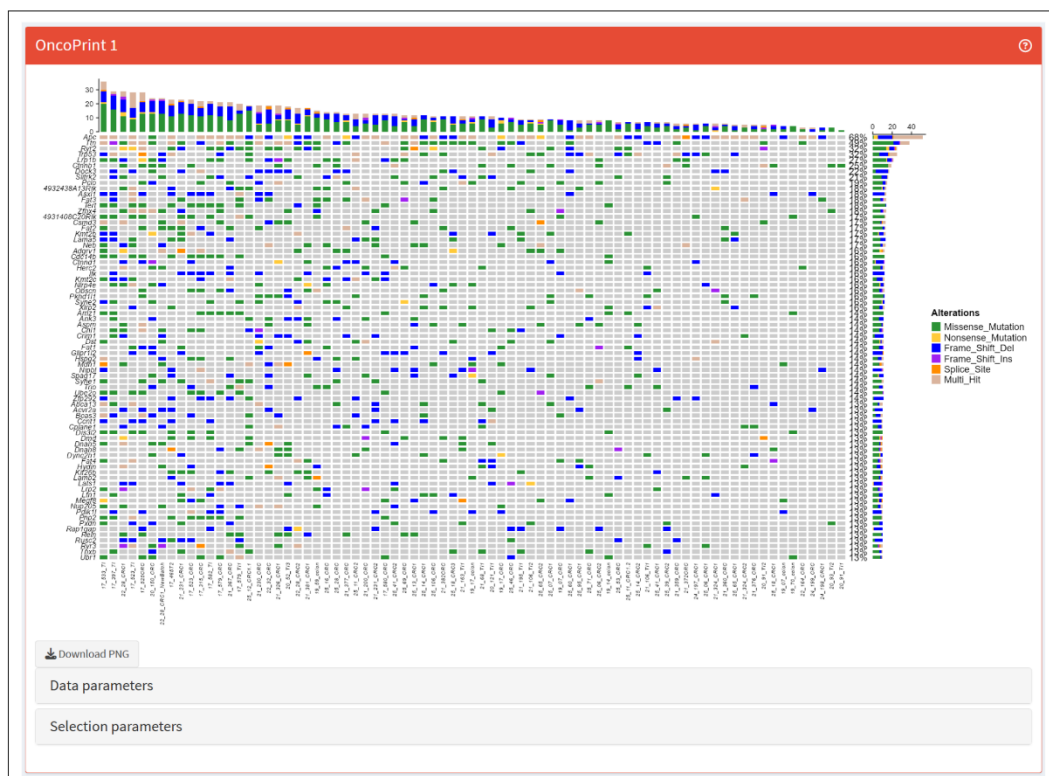


Figure S3. OncoPlot panel showing the alteration matrix for the tumor cohort, with genes ordered by mutation frequency (top bar plot) and gene mutation frequencies shown in the right-hand bar plot. Alteration types (missense mutation, nonsense mutation, frame shift deletion/insertion, splice site, multi-hit) are indicated by color.

1.3 Panel: ColumnDataTable

This panel is only available when sample metadata has been uploaded. It shows the metadata table (one row per sample) with a search box for each column. Typing in the search box filters which samples are shown, and this selection is automatically passed to the OncoPlot panel, so the OncoPlot updates to show only the matching samples.

What you can change:

- Search box (one per column): type a value to filter the samples shown, and to restrict the samples drawn in the OncoPlot.

Column data table 1

Show 10 entries Search:

	Treatment.groups	Tumor.vs.Healthy
	All	All
25_12_CRC1.1	MSH2Braf	CRC
17_579_CRC	MSH2	CRC
25_69_CRC	MSH2Braf	CRC
25_60_CRC1	MSH2Braf	CRC
17_465T2	MSH2	SI tumor
17_523_TI	MSH2	SI tumor
25_106_CRC	MSH2Braf	CRC
22_28_CRC2	MSH2Braf	CRC
21_367_CRC	MSH2	CRC
25_13_CRC1	MSH2Braf	CRC

Showing 1 to 10 of 77 entries Previous 1 2 3 4 5 ... 8 Next

Data parameters

Selection parameters

Active selection shown below table (if any).
Transmitting selection to *OncoPrint 1*

Figure S4. ColumnDataTable panel of the Mutations application, showing the sample metadata table (77 samples in total, Treatment.groups and Tumor.vs.Healthy columns)

2. RNA-seq application

The Bulk RNA-seq application explores gene expression data. The user uploads a single DeeDeeExperiment object (.rds file) containing expression values, and optionally the results of a differential expression analysis (DEA) and a functional enrichment analysis (FEA) for one or more contrasts. The application has two tabs: Basic Exploration and DEA and FEA results.

2.1 Sidebar controls

These controls appear in the sidebar and affect several panels at once. Some only appear on one of the two tabs.

- Upload .rds data: the DeeDeeExperiment file to explore.
- Clustering by: which sample metadata column is used to color samples and group them in the dendrogram, PCA and heatmap.
- Number of genes to display (Basic Exploration tab only): how many of the most variable genes are used for the heatmap and gene selector.
- Select samples: which samples to include in the plots.
- Select genes: restricts plots to specific genes of interest.
- Select Contrast for DEA (DEA and FEA results tab only): which comparison(s) to look at. Choosing 2-4 contrasts also enables the Venn diagram.
- P adjusted Value threshold (DEA and FEA results tab only): significance cut-off used by the volcano plot, Venn diagram, KaryoPlot and CircosPlot.
- Log2 Fold Change threshold (DEA and FEA results tab only): fold-change cut-off used by the same panels.
- Zoom to chromosome (DEA and FEA results tab only, if genomic position is available): which chromosome is shown in the zoomed KaryoPlot.

The sidebar of the RNA-seq application is divided into several sections:

- Input data:** Includes a section for uploading .rds data (DeeDeeExperiment) with a 'Browse...' button and a file named 'HPalmer_MSH2BR'. Below this is an 'Upload complete' button.
- Parameters:** Contains a 'Clustering by:' dropdown menu set to 'Sample'.
- Number of genes to display (min 2 - max 2000):** A text input field set to '50'.
- Select samples to visualize:** A list of selected samples: '17-315_CRC' and '17-525_CRC', each with a close button (x).
- Select top variable genes to visualize:** A text input field labeled 'Select genes...'.
- Select Contrast for DEA:** A list of selected contrasts: 'Msh2Braf.vs.WT', 'Msh2.vs.WT', and 'Msh2Braf.vs.Msh2', each with a close button (x).
- P adjusted Value Threshold:** A text input field set to '0,05'.
- Log2 Fold Change Threshold:** A text input field set to '1'.
- Zoom to chromosome:** A dropdown menu set to 'chr1'.

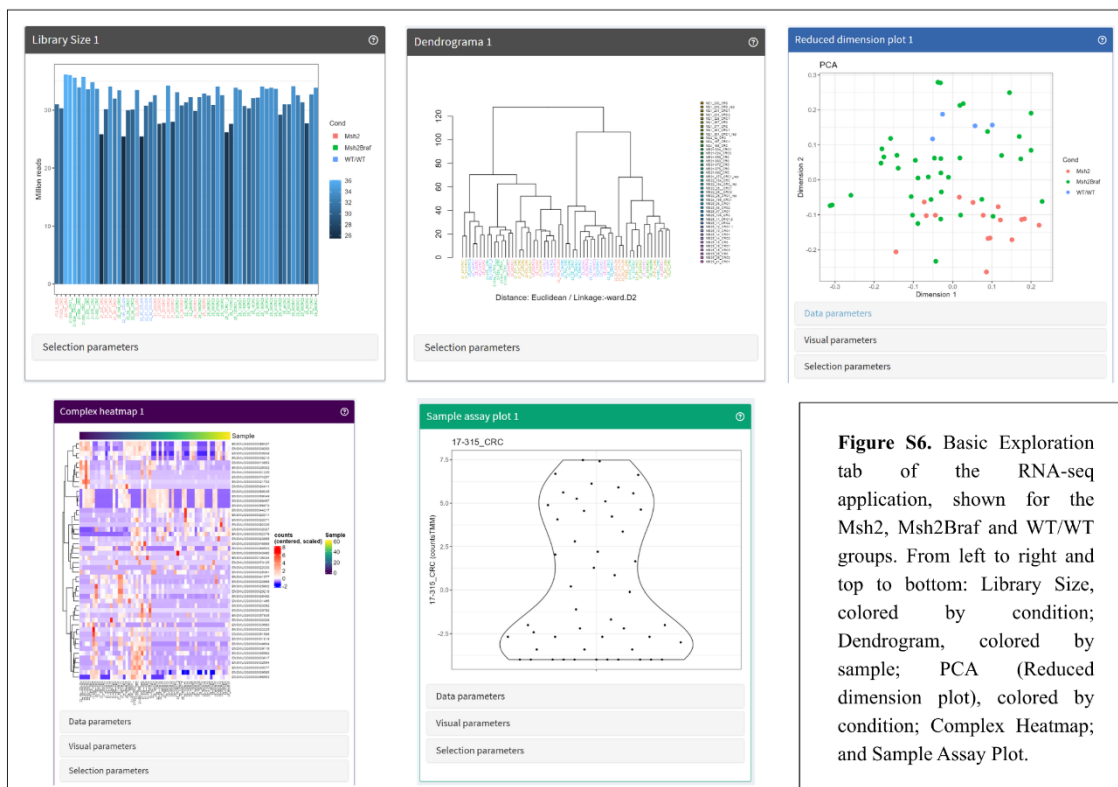
Figure S5. Sidebar of the RNA-seq application, showing the required inputs, filters described.

2.2 Panels: Basic Exploration Tab

This tab is for quality control and for getting a first look at how samples group together, before any statistical testing. The panels are linked: selecting samples in the PCA plot updates the Library Size, Dendrogram and Heatmap panels to show only those samples.

Panel	What it shows	What you can change
Library Size	Bar plot of the total number of reads (millions) per sample, colored by the chosen grouping variable. Quick check for uneven sequencing depth.	Reacts to Select samples and Clustering by (sidebar).
Dendrogram	Hierarchical clustering tree (Euclidean distance, Ward.D2) showing how similar samples are to each other, colored by the chosen grouping variable.	Reacts to Select samples, Select genes, Number of genes to display, and Clustering by (sidebar)
PCA	Principal Component Analysis plot; samples with similar	Reacts to Select samples and Clustering by (sidebar); standard

	overall expression appear close together, colored by the chosen grouping variable.	iSEE controls for which components to plot (X/Y axis), point size and color options (panel settings).
Heatmap	Expression level of the most variable genes across samples, scaled by row and annotated by group.	Number of genes to display and Select genes (sidebar); clustering and color-scale options (panel settings).
SampleAssayPlot	Expression value of one selected gene across all samples.	Which gene to display, and coloring/grouping options (panel settings).

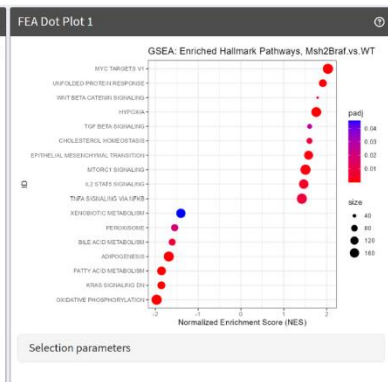
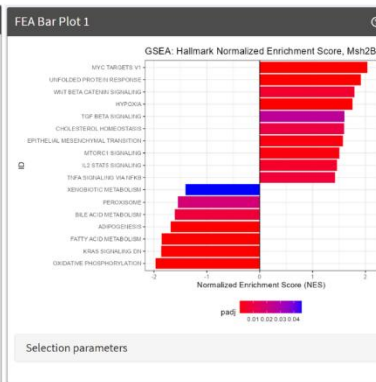
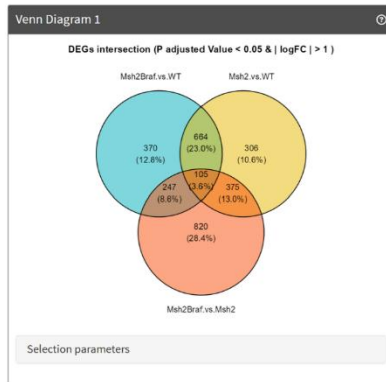
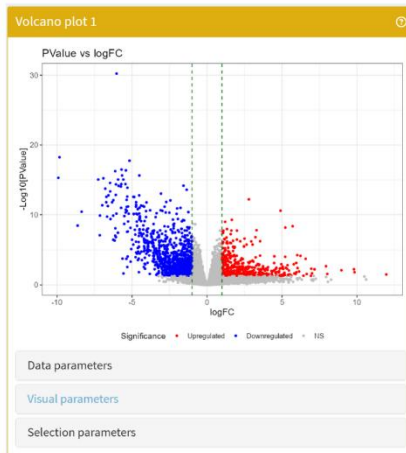
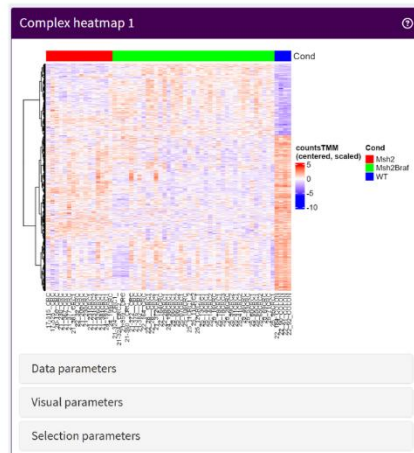


2.3 Panels: DEA and FEA results Tab

This tab shows the results of the differential expression analysis (DEA) and functional enrichment analysis (FEA) for the contrast(s) selected in the sidebar. Panels are linked: clicking a gene in the RowDataTable highlights it on the KaryoPlot panels, and filtering for up- or down-regulated genes highlights those genes on the Volcano Plot.

Panel	What it shows	What you can change
Volcano Plot	Every gene plotted by fold change (X axis) and significance (Y axis) for the first selected contrast.	P adjusted Value and Log2 Fold Change thresholds (sidebar); point size, color and label options (panel settings).
RowDataTable (gene table)	Searchable table with one row per gene: gene annotation plus DEA statistics (log2FoldChange	Search box per column; clicking a row selects a gene, highlighted in red on the KaryoPlot panels.

	and adjusted p-value) for the first selected contrast..	
ColumnDataTable (sample table)	Searchable table with one row per sample, showing sample metadata.	Search box per column; selection restricts which samples appear in the Heatmap.
Heatmap	Expression of the differentially expressed genes (DEGs) for the selected contrast, samples annotated by group.	Select genes (sidebar), clustering and color-scale options (panel settings).
Venn Diagram	Overlap of significant genes between 2-4 selected contrasts. Only appears when 2 or more contrasts are chosen.	Select Contrast for DEA (2-4); P adjusted Value and Log2 Fold Change thresholds (sidebar).
FEA Dot Plot	Top enriched Hallmark pathways (GSEA) for the first selected contrast; dot position = NES, size = gene count, color = adjusted p-value.	No settings of its own; determined by the first selected contrast. The pathway significance cutoff ($\text{padj} < 0.05$) is fixed in the code.
FEA Bar Plot	Same enrichment results as the FEA Dot Plot, shown as a bar chart for easier comparison of direction and strength.	No settings of its own; determined by the first selected contrast. The pathway significance cutoff ($\text{padj} < 0.05$) is fixed in the code.
KaryoPlot (whole genome)	All chromosomes side by side, marking where the DEGs of each selected contrast are located across the genome.	Select Contrast for DEA, P adjusted Value and Log2 Fold Change thresholds (sidebar). Gene selection in RowDataTable.
KaryoPlot (zoom)	Same as the whole-genome KaryoPlot, restricted to a single chromosome.	Zoom to chromosome (sidebar), plus the same contrast/threshold controls. Gene selection in RowDataTable.
CircosPlot	DEGs of two contrasts shown in a circular layout, up- and down-regulated genes in different colors, for direct visual comparison.	Contrast 1 / Contrast 2 (panel settings, set from the first two selected contrasts); P adjusted Value and Log2 Fold Change thresholds (sidebar).



Row data table 1

Show 10 entries

Search:

Geneid	SYMBOL	Length	Description	Msh2Braf.vs.WT_log2FoldChange	Msh2Braf.vs.WT_padj	Significance	rowRanges_star
All	A		All	All	All	All	All
ENSMUSG000000000001	Gnai3	38866	G protein subunit alpha i3	0.0223907807229109	0.902927872330842	NS	108107
ENSMUSG000000000028	Cdc45	31540	cell division cycle 45	-0.231759250538696	0.669020302516447	NS	18780
ENSMUSG000000000031	H19	2614	H19, imprinted maternally expressed transcript	2.99519004983492	0.188586249491398	NS	142575
ENSMUSG000000000037	Scml2	141020	Scm polycomb group protein like 2	-2.44343451960153	0.00295498597233358	Downregulated	161117

Column data table 1

Show10entries

Search:

Sample	Cond	Tissue	Batch	
All	All	All	All	
17-315_CRC	M17-315_CRC	Msh2	CRC	Batch1
17-525__CRC	M17-525_CRC	Msh2	CRC	Batch1
20-150_CRC	M20-150_CRC	Msh2	CRC	Batch1
21-324__CRC1_	MB21-324_CRC1	Msh2Braf	CRC	Batch1
21-324__CRC2____	MB21-324_CRC2	Msh2Braf	CRC	Batch1
21-359__CRC	MB21-359_CRC	Msh2Braf	CRC	Batch1

Figure S7. Panels of the DEA and FEA results tab, shown for the three contrasts (Msh2Braf.vs.WT, Msh2.vs.WT and Msh2Braf.vs.Msh2), using an adjusted p-value < 0.05 and |log fold change| > 1. Complex heatmap shows the expression (centered and scaled counts) of the DEGs, with samples annotated by condition. Volcano plot shows all genes plotted by log fold change and significance for the first selected contrast (Msh2Braf.vs.WT), colored as Upregulated, Downregulated or Not Significant. Venn Diagram shows the overlap of significant genes between the three contrasts. FEA Bar Plot and FEA Dot Plot show the top enriched Hallmark pathways (GSEA) for the first selected contrast. RowDataTable shows the gene-level table with annotation and DEA statistics for the first selected contrast. ColumnDataTable shows the sample metadata table.



Figure S8. KaryoPlot and CircosPlot panels of the DEA and FEA results tab, shown for the three contrasts (Msh2Braf.vs.WT, Msh2.vs.WT and Msh2Braf.vs.Msh2), using an adjusted p-value < 0.05 and |log fold change| > 1. Genomic Karyoplot 1 shows the DEGs of the three contrasts across all chromosomes, with the gene *Pkhd1* highlighted in red following its selection in the RowDataTable. Genomic Karyoplot 2 shows the same data zoomed to chromosome 1, with *Pkhd1* again highlighted in red. Circos Plot compares the DEGs of two of the three contrasts (Msh2Braf.vs.WT and Msh2.vs.WT), with upregulated and downregulated genes shown in different colors for each contrast..